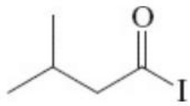
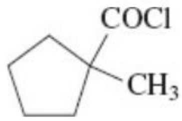
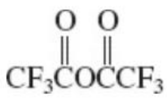
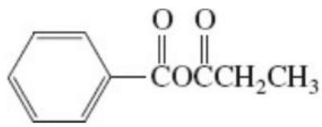
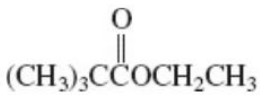
30. Name (IUPAC system) or draw the structure of each of the following molecules.

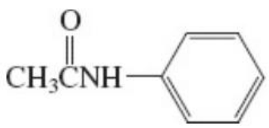
a. 

b. 

c. 

d. 

e. 

f. 

g. Propyl butanoate

h. Butyl propanoate

i. 2-Chloroethyl benzoate

j. *N*,*N*-Dimethylbenzamide

k. 2-Methylhexanenitrile

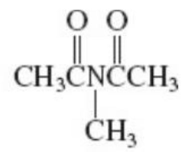
l. Cyclopentanecarbonitrile

31. Name each of the following structures according to IUPAC or *Chemical Abstracts* rules. Pay attention to the order of precedence of the functional groups.

32. **(a)** Use resonance forms to explain in detail the relative order of acidity of carboxylic acid derivatives, as presented in Section 20-1. **(b)** Do the same, but use an argument based on inductive effects.

33. In each of the following pairs of compounds, decide which possesses the indicated property to the greater degree. **(a)** Length of C–X$\mathrm{C{-}X}$ bond: acetyl fluoride or acetyl chloride. **(b)** Acidity of the boldface H: CH2(COCH3)2$\mathbf{CH_2}(COCH_3)_2$ or CH2(COOCH3)2.$\mathbf{CH_2}(COOCH_3)_2$. **(c)** Reactivity toward addition of a nucleophile: (i) an amide or (ii) an imide (as shown below). **(d)** High-energy infrared carbonyl stretching frequency: ethyl acetate or ethenyl acetate.

i



ii

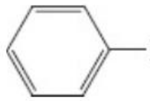
34. Give the product(s) of each of the following reactions.

35. Formulate a mechanism for the reaction of acetyl chloride with 1-propanol shown on p. 975.

36. Give the product(s) of the reactions of acetic anhydride with each of the following reagents. Assume in all cases that the reagent is present in excess.

a. (CH3)2CHOH$(CH_3)_2CHOH$

b. NH3NH_3

c. MgBr, THF; then H^+, H_2O

37. Give the product(s) of the reaction of butanedioic (succinic) anhydride with each of the reagents in Problem 36.

38. Formulate a mechanism for the reaction of butanedioic (succinic) anhydride with methanol shown on p. 978.

39. Give the products of reaction of methyl pentanoate with each of the following reagents under the conditions shown.

a. NaOH, H2O,H_2O, heat; then H+,H2OH^+, H_2O

b. (CH3)2CHCH2CH2OH$(CH_3)_2CHCH_2CH_2OH$ (excess), H+H^+

c. (CH3CH2)2NH,$(CH_3CH_2)_2NH$, heat

d. CH3MgICH_3MgI (excess), (CH3CH2)2O;$(CH_3CH_2)_2O$; then H+,H2O H^+, H_2O

e. LiAlH4,(CH3CH2)2O;$LiAlH_4, (CH_3CH_2)_2O$; then H+,H2OH^+, H_2O

f. [(CH3)2CHCH2]2AlH,$[(CH_3)_2CHCH_2]_2AlH$, methylbenzene (toluene), low temperature; then H+,H2OH^+, H_2O

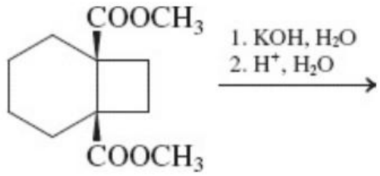
40. Give the products of reaction of γγ-valerolactone (5-methyloxa- 2-cyclopentanone, Section 20-4) with each of the reagents in Problem 39.

42. Formulate a mechanism for the acid-catalyzed transesterification of ethyl 2-methylpropanoate (ethyl isobutyrate) into the corresponding methyl ester. Your mechanism should clearly illustrate the catalytic role of the proton.

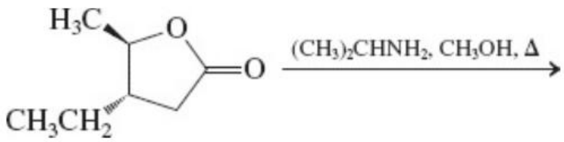
44. Reaction review. Suggest reagents to convert each of the following starting materials into the indicated product: **(a)** acetyl chloride into acetic hexanoic anhydride; **(b)** methyl hexanoate into *N*-methylhexanamide; **(c)** hexanoyl chloride into hexanal; **(d)** hexanenitrile into hexanoic acid; **(e)** hexanamide into hexanamine; **(f)** hexanamide into pentanamine; **(g)** ethyl hexanoate into 3-ethyl-3-octanol; **(h)** hexanenitrile into 1-phenyl-1-hexanone [C6H5CO(CH2)4CH3]. $[C_6H_5CO(CH_2)_4CH_3]$.

45. Give the product of each of the following reactions.

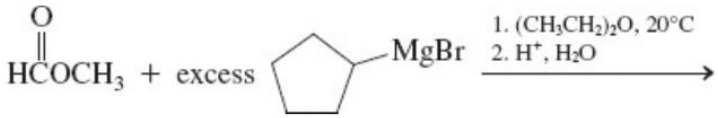
a.



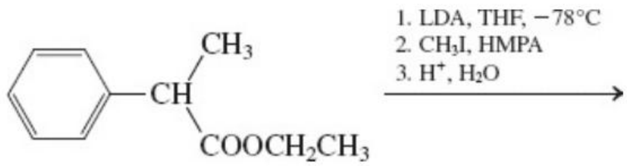
b.



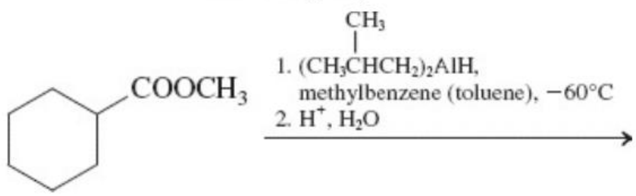
c.



d.

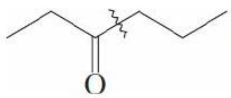


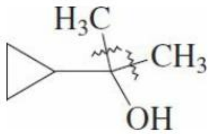
e.



52. What reagents would be necessary to carry out the following transformations? **(a)** Cyclohexanecarbonyl chloride →→ 1-cyclohexyl-1-pentanone; **(b)** 2-butenedioic (maleic) anhydride →→ (*Z*)-2-butene-1,4-diol; **(c)** 3-methylbutanoyl bromide →→ 3-methylbutanal; **(d)** benzamide →→ 1-phenylmethanamine; **(e)** propanenitrile →→ 3-hexanone; **(f)** methyl propanoate →→ 4-ethyl-4-heptanol.

53. In each of the following retrosynthetic disconnections (∿), show reactions that would make the indicated C−CC−C bonds.

a.  (Two reactions)

b. 

c.